

Analyzing Classical Machine Learning vs Deep Learning for Sentiment Analysis

Abstract

This project investigates the comparative performance of classical machine learning and deep learning models for sentiment analysis using the Yelp review dataset. The goal is to determine whether deep learning methods such as LSTM and TinyBERT consistently outperform simpler models like Logistic Regression and Linear SVM, especially when the amount of training data is limited. The Yelp dataset contains user reviews paired with star ratings, which were converted into binary sentiment labels for positive (4–5 stars) and negative (1–2 stars) classes. To evaluate scalability, two subsets were created (20,000 and 50,000 reviews). Classical models were trained using TF-IDF vectorization, while LSTM and TinyBERT relied on embedding and transformer based representations to capture contextual information. All models were evaluated using Accuracy, Precision, Recall, F1-score, and ROC-AUC to ensure a comprehensive comparison of performance. Results show that Logistic Regression and Linear SVM demonstrate strong and stable performance across both dataset sizes, achieving accuracies between 93.51%–95.07% and F1-scores between 95.88%–96.88%, highlighting their effectiveness and computational efficiency. LSTM showed weaker performance on the small dataset (Accuracy = 88.37%, F1 = 92.77%) but significantly improved on the large dataset (Accuracy = 94.96%, F1 = 96.82%), suggesting that deep learning models benefit substantially from increased training data. However, LSTM required considerably longer training times (470–1042 seconds), which introduces practical constraints. TinyBERT, adopted instead of full BERT or DistilBERT due to computational limitations, delivered the strongest overall deep learning performance. On the small dataset, TinyBERT achieved 93.87% accuracy (F1 = 96.19%), outperforming LSTM. On the large dataset, it achieved the highest accuracy (95.44%), the highest F1-score (97.11%), and the highest ROC-AUC (98.67%), though with substantially higher training times (3092–7753 seconds). These results show that classical models remain dependable and efficient baselines, while lightweight transformer models like TinyBERT offer superior accuracy when computational cost is acceptable.

1. Introduction

Sentiment analysis is a widely used technique for determining whether a given piece of text expresses a positive or negative opinion, with applications in areas such as product reviewing, social media monitoring, customer satisfaction analysis, and business intelligence. As online platforms continue generating large volumes of user generated text, developing accurate and scalable sentiment classification models has become increasingly important. This project aims to compare classical machine learning models Logistic Regression and Support Vector Machines (SVM) with deep learning models such as Long Short-Term Memory networks (LSTM) and TinyBERT for sentiment classification on the Yelp review dataset. The central question is whether deep learning methods justify their higher computational cost and architectural complexity by offering significantly improved performance, particularly as dataset size increases. The Yelp dataset includes millions of user reviews paired with star ratings, which were converted into binary sentiment labels to simplify the analysis. To evaluate generalizability and scalability, two subsets were created: a small dataset containing 20,000 reviews and a large dataset containing 50,000 reviews. Classical models were trained using TF-IDF vectorization to represent text features, whereas the LSTM model used embedding based representations to capture sequential dependencies and contextual meaning. TinyBERT, a lightweight transformer model distilled from BERT, was included to explore whether transformer based architectures can deliver stronger semantic understanding while remaining feasible under computational constraints. TinyBERT was selected specifically because full BERT and DistilBERT were too resource intensive for the available hardware. Preliminary results show that classical models outperform LSTM on the smaller dataset, achieving higher accuracy and F1-scores while requiring significantly less computation time. LSTM performance improves substantially with more data, indicating that deep learning architectures are highly data dependent but capable of learning richer semantic patterns. TinyBERT provides a middle ground between classical models and heavier transformer architectures: on the small dataset, it performs comparably to Logistic Regression and SVM while delivering stronger contextual understanding, and on

the large dataset it achieves one of the highest overall accuracies and F1-scores in the experiment. These results show that transformer based models can scale effectively with increased data while remaining computationally manageable, offering a valuable balance between accuracy, scalability, and efficiency.

2. Background/Related Work:

2.1. Logistic Regression and SVM:

Logistic regression is a widely used method that has been in development since the early 19th century and gained popularity for classification tasks in the later 1900s [1]. It is simple, computationally efficient, and effective on linearly separable data, but can struggle with non linear relationships or class imbalance. Support Vector Machines (SVM) can address some of these limitations as they allow for non linear kernels and work well with unstructured data like text or images. However, their computational requirements “increase rapidly with the number of training vectors” [2]. Support Vector Classification (SVC) may have up to quadratic time complexity, while a linear kernel implementation (LinearSVC in Python) can improve computational efficiency and scale more effectively to large datasets. Despite the strengths of these classical machine learning methods, both can struggle to capture contextual and sequential patterns in text for sentiment analysis.

2.2. Long Short Term Memory:

Deep learning, inspired by the hierarchical structure of the human brain, gained popularity in the early 2010s. These models automatically learn features from raw text and often outperform traditional models on large datasets. However, they typically require a significant amount of training data, computational resources, and do not produce interpretable parameters which makes it difficult to identify the impact of specific features on the prediction. Recurrent Neural Networks (RNNs) are a type of neural network that are trained on sequential data and perform well on identifying patterns in text. They differ from simple applications of feedforward networks, such as Multi Layer Perceptron (MLP), due to their cyclical nature and their ability to “take into account previous inputs and not only the current input” [3]. However, they struggle to retain information from many steps back and can be impacted by vanishing and exploding gradients. Hochreiter and Schmidhuber introduced a gradient based method to mitigate these issues in their 1997 paper “Long Short Term Memory,” which gained significant popularity and has remained as a widely used approach to this day [4]. LSTMs use a gated approach to control the amount of memory retained, forgotten, and passed to subsequent time steps during training. They are strong in their ability to bridge information between long term depen-

dencies, reduce the vanishing gradient problem experienced by RNNs, and are scalable for large datasets [4].

2.3. Bidirectional Encoder Representations from Transformers:

The approaches highlighted above have been widely used since their creation, but a major breakthrough in deep learning came in 2017 with Vaswani et al.’s paper “Attention Is All You Need” [5]. The paper proposed the Transformer model architecture, which avoids “recurrence and instead relies entirely on an attention mechanism to draw global dependencies between input and output” [5]. The encoder-decoder style architecture allows the full input sequence to be processed at once, which provides computational efficiency as well as a self attention mechanism which gains context between each word in the sentence. Additionally, it uses positional encodings to learn sentence structure and multi-head attention that provides enhanced contextual understanding. This powerful architecture provided the foundation for many modern advanced models, such as Bidirectional Encoder Representations from Transformers (BERT) in 2018. BERT is designed to train “deep bidirectional representations from unlabeled text by jointly conditioning on both left and right context in all layers” [6]. Although the architecture can learn strong contextual understanding in sentiment analysis, it has a high computational cost and is prone to overfitting. To reduce the computational complexity while maintaining accuracy, Jiao et al. proposed a “novel Transformer distillation method” in the 2019 paper “TinyBERT: Distilling BERT for Natural Language Understanding” [7]. The architecture uses a two stage distillation process to reduce the original BERT model into a significantly smaller model while preserving its performance.

2.4. Related Work:

A previous study, “Sentiment Analysis of Yelp Reviews: A Comparison of Techniques and Models” by Siqi Liu, compared classical approaches such as Logistic Regression and SVM with deep learning models like LSTM and BERT on the Yelp dataset [8]. The author found that simpler models often achieved comparable or better performance when training data was limited. Similarly, “Evaluation of Machine Learning Approach for Sentiment Analysis using Yelp Dataset” by Sadikin et. al evaluated an LSTM model on Yelp restaurant reviews and demonstrated its strength in capturing contextual meaning but did not include direct comparisons with classical baselines [9]. This project differs from these prior studies by performing a systematic, controlled comparison of Logistic Regression, Linear SVM, LSTM, and BERT across two dataset scales (20k and 50k) under identical preprocessing and evaluation criteria. This dual scale approach allows explicit analysis of how dataset size affects model performance and generalization, offering

a more balanced perspective on when deep learning methods truly outperform classical models in real world sentiment analysis tasks.

3. Approach:

The project compares performance between classical machine learning and deep learning approaches in training time and performance metrics (Accuracy, Precision, Recall, F1-score, and ROC-AUC) for a sentiment analysis classification task. The experiment performs a systematic, controlled comparison of the models by applying identical datasets, preprocessing, and evaluation across all candidates.

3.1. Hypotheses:

1. Classical models (Logistic Regression and Linear SVC) will achieve comparable or better performance than deep learning models (LSTM and TinyBERT) on the small dataset (20k reviews).
2. Deep learning models (LSTM and especially TinyBERT) will outperform classical models on the larger dataset (50k reviews), due to their ability to learn richer semantic and contextual relationships.
3. Deep learning models will require substantially higher training time and computational resources than classical models across both datasets.

3.2. Data Preprocessing:

The Yelp reviews dataset was chosen since it is representative of realistic sentiment, since it was taken from actual reviews on the Yelp platform. The first step involved importing the “yelp_academic_dataset_review.json” file, which was downloaded directly from the Yelp website and uploaded to Google Drive so that the Colab environment could retrieve the file with a mount. Given the dataset’s extensive size (more than 300 million reviews), only the first 200,000 samples were retrieved to build the small and large datasets. Reviews with 3 stars were removed and the star ratings were binary encoded (4 and 5 star ratings = 1, 1 and 2 star ratings = 0) to simplify the classification task. The small and large datasets (20k and 50k samples respectively) were randomly selected from the remaining samples for further analysis. The class distributions of both datasets were analyzed and both included around 80% positive and 20% negative reviews. Although the class distributions were imbalanced, the distribution was retained to match real world applications, and because several evaluation metrics (precision, recall, ROC-AUC) are not impacted by class imbalance. The small and large datasets were split into a 70/15/15 distribution using the `train_test_split` package from `scikit-learn`.

For Logistic Regression and SVM, the inputs were vectorized using the `TfidfVectorizer` package from `scikit-learn` with `max_features = 5000` to limit the vocabulary size. For LSTM, the inputs were tokenized using the `Tokenizer` package from the TensorFlow Keras library and `max_words = 5000` to remain consistent with the classical machine learning model runs. For the TinyBERT run, the reviews were tokenized using the `AutoTokenizer` package from the `transformers` library using a pretrained model. Reviews were padded to a maximum length of 200 words for all models to maintain consistency, which covered 90% of the dataset and to reduce computational cost.

3.3. Model Training:

The first model implemented was Logistic Regression using `scikit-learn` library. Separate models were trained on the small and large training sets for direct performance comparison. No regularization penalties were applied to ensure comparability between the proposed models and since limiting vocabulary size already reduced complexity. The model was limited to 2000 maximum iterations to prevent overfitting. The second classical machine learning approach used was a Linear Support Vector Classification (SVC) model, which used the `LinearSVC` package from `scikit-learn`. Similar to the approach in Logistic Regression, two separate models were trained on the small and large training datasets. Both models used 2000 maximum iterations and `C = 1.0`, the default value in `scikit-learn`. The first deep learning candidate model was the Long Short-Term Memory (LSTM) neural network. The sequential, embedding, LSTM, dense, and dropout packages from TensorFlow were used to build the model architectures for the small and large dataset approaches. The architectures both included one embedding layer, two LSTM layers, two dropout layers with dropout rate = 0.2, and one dense output layer that used a sigmoid activation function. The hidden embedding layers used 64 dimensions each, the adam optimizer was used to reduce computational time, and each model was run for 5 epochs. The final model that was trained was TinyBERT, which was selected instead of BERT or DistilBERT due to computational constraints. The reviews were organized into PyTorch datasets and dataloaders for batching with the help of ChatGPT. The specific TinyBERT model was the “huawei-noah/TinyBERT_General_4L_312D”, which is a smaller version of BERT with 4 transformer layers and 312 hidden dimensions. The model was trained for 2 epochs using the AdamW optimizer separately on the small and large datasets for later comparison.

4. Experiment:

In this experiment, four models Logistic Regression, Linear SVM, LSTM, and TinyBERT were trained and evaluated on two subsets of the Yelp reviews dataset (20k

and 50k samples). Each model used identical 70/15/15 train-validation-test splits to ensure a controlled comparison across all approaches. Classical models were trained on TF-IDF features, while LSTM and TinyBERT were trained using tokenized text padded to a fixed sequence length. After training, predictions were generated on the validation set, and model performance was measured using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Training times were recorded using Python’s time library to compare computational efficiency between classical and deep learning models. Results were reported through detailed tables and visualizations, including training time comparisons, accuracy differences between small and large datasets, and F1-score vs. training time trade offs. TinyBERT was included to examine whether transformer-based architectures can outperform traditional and recurrent models under limited computational resources. Qualitative analysis also focused on how dataset size affects each model’s ability to learn meaningful sentiment patterns.

Table 1. Performance of Logistic Regression on Small and Large Yelp Datasets.

Metric	Small Dataset	Large Dataset
Accuracy (%)	93.90	93.51
Precision (%)	95.92	95.90
Recall (%)	96.37	95.85
F1-Score (%)	96.14	95.88
ROC-AUC (%)	97.17	97.46
Training Time (s)	0.46	1.52

Table 2. Performance of Linear SVM on Small and Large Yelp Datasets.

Metric	Small Dataset	Large Dataset
Accuracy (%)	94.47	95.07
Precision (%)	95.99	96.54
Recall (%)	97.04	97.23
F1-Score (%)	96.51	96.88
ROC-AUC (%)	90.95	92.14
Training Time (s)	0.33	1.20

Table 3. Performance of LSTM on Small and Large Yelp Datasets.

Metric	Small Dataset	Large Dataset
Accuracy (%)	88.37	94.96
Precision (%)	91.01	96.30
Recall (%)	94.59	97.34
F1-Score (%)	92.77	96.82
ROC-AUC (%)	79.87	91.73
Training Time (s)	470.35	1042.87

Table 4. Performance of TinyBERT on Small and Large Yelp Datasets.

Metric	Small Dataset	Large Dataset
Accuracy (%)	93.87	95.44
Precision (%)	94.24	97.14
Recall (%)	98.22	97.07
F1-Score (%)	96.19	97.11
ROC-AUC (%)	98.21	98.67
Training Time (s)	3092.32	7752.69

As shown above in Tables 1-4, the classical models Logistic Regression and Linear SVM delivered consistently strong performance, with F1-scores around 96% and accuracies between 93–95%. Their results were stable across both dataset sizes, and training times remained under one second, making them highly efficient and reliable baselines for sentiment analysis. In contrast, the LSTM model showed clear sensitivity to dataset size: its performance dropped notably on the small dataset (F1 = 92.77%, ROC-AUC = 79.87%) but improved significantly with more data, reaching competitive results on the large dataset. However, this improvement came with substantial computational cost, requiring hundreds to thousands of seconds to train. TinyBERT provided the most balanced performance among the deep learning models. Even on the small dataset, it achieved strong accuracy (93.87%) and F1-score (96.19%), outperforming LSTM under limited data. On the large dataset, TinyBERT delivered the highest overall scores including 95.44% accuracy and 98.67% ROC-AUC demonstrating the advantage of transformer based contextual representations. Although TinyBERT required more computation than the classical models, its performance gains were significant.

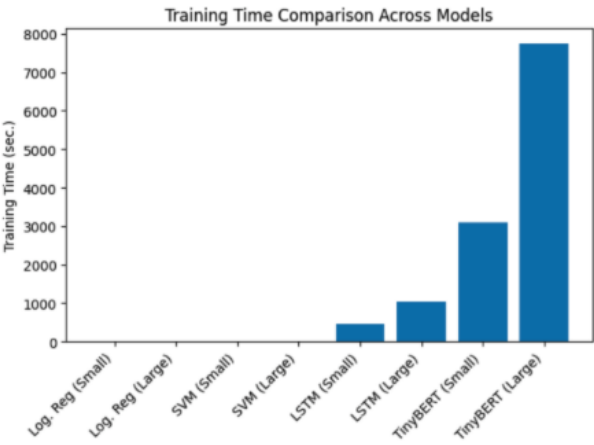


Figure 1. Training time comparison across all candidate models.

Figure 1 visualizes the substantial increases in training

time for the deep learning models. Classical models trained in minute time (less than 2 seconds) compared to the more complex models. LSTM required hundreds to over one thousand seconds, and TinyBERT required over 7,700 seconds on the large dataset. This contrast emphasizes the steep computational cost of deep learning models relative to the classical approaches.

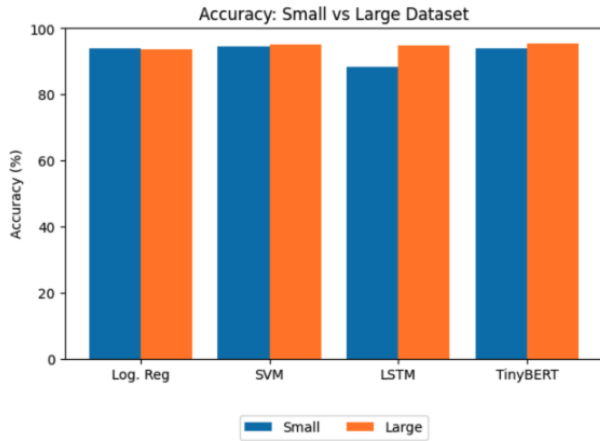


Figure 2. Accuracy comparison between small and large datasets across all candidate models.

Figure 2 shows that accuracy remained stable for the classical models and TinyBERT, since there were not large disparities between the accuracies on the small and large datasets. This demonstrates that the models are robust to sample size. However, LSTM improved substantially when trained on the large dataset, increasing from 88% to nearly 95%. This indicates that the model relies more heavily on training information to learn semantic patterns compared to the other candidate models.

Figure 3 illustrates the trade off between training time and F1-score, showing that classical models achieve high performance with minimal computation, while TinyBERT delivers the strongest F1-score but at a substantially higher training cost. Overall, the results across all figures highlight the trade offs between efficiency and accuracy: classical models remain excellent low cost options, LSTM benefits from large datasets but is computationally expensive, and TinyBERT offers near state of the art performance while remaining significantly lighter than full scale BERT models.

5. Conclusion:

This project compared classical machine learning models (Logistic Regression and Linear SVM) with deep learning approaches (LSTM and TinyBERT) for sentiment analysis on two differently sized subsets of the Yelp review dataset. The results highlight a clear pattern: classical models remain highly competitive, especially on smaller

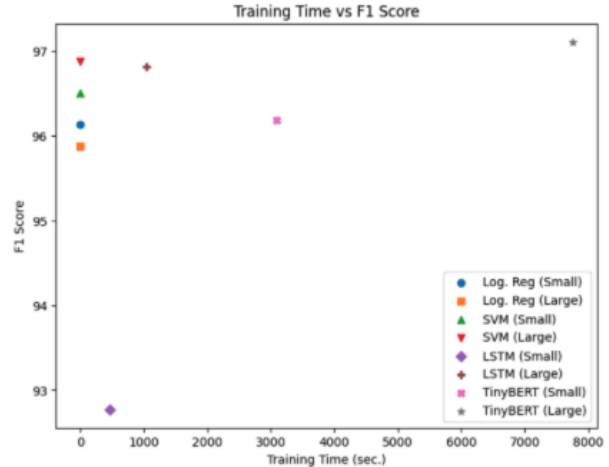


Figure 3. Comparison of training time (sec.) vs. F1 score across models and dataset size.

datasets, offering strong accuracy and F1-scores with minimal computational cost. Their speed and stability make them reliable baselines when resources or training time are limited. LSTM showed strong improvement as dataset size increased, reinforcing the idea that recurrent neural networks depend heavily on larger training samples to capture meaningful sequential patterns. While its performance eventually approached that of classical models, the significantly higher training times make LSTM less practical unless substantial data and resources are available. TinyBERT delivered the most notable gains. It matched classical model performance on the smaller dataset and achieved the highest accuracy, F1-score, and ROC-AUC on the larger dataset. These results illustrate the strength of transformer based architectures, even in distilled and lightweight form, and highlight the benefit of leveraging pretrained contextual embeddings. However, TinyBERT's long training times show that even distilled transformers come with considerable computational overhead. Overall, this project demonstrates that the best model depends on the balance of accuracy needs, dataset size, and available hardware. Classical models remain efficient and reliable, while transformers offer superior performance when resources allow. There are several promising directions for extending this analysis. TinyBERT could be trained for additional epochs using stronger computational resources to determine whether its performance continues to improve with deeper fine tuning. Although TinyBERT already outperformed other models on the larger dataset, longer training may further enhance its contextual understanding. Future work could also involve fine tuning the full BERT model or DistilBERT, which was not feasible in this project due to hardware limitations, to compare how much performance gain full scale transformers provide over distilled versions. Additionally, the cur-

rent binary labeling scheme excluded 3 star reviews. Incorporating these into the dataset as a neutral class or and performing multi class classification could provide a more realistic evaluation of the models. Finally, expanding the dataset beyond 50,000 samples and experimenting with a substantially larger corpus would help evaluate how model performance scales under more demanding conditions.

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